

SOME PHYSICS STUDIES IN SUPPORT OF PFR DESIGN

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1. Introduction

This paper discusses some of the physics problems which require solution during the design and early commissioning stages of the PFR reactor. A complementary paper⁽¹⁾ is concerned more generally with the methods used in making predictions of physics parameters of large fast reactors.

2. Nuclear Data and Calculation Methods Used for the PFR

For the PFR, in which the neutron spectrum is relatively degraded, many cross-section sets are unsatisfactory either through inadequate low energy representation, absence of resonance self-shielding or inadequate high energy group structure. The majority of calculations referred to in this paper have used the data set FD2⁽²⁾ which has been developed for use in large sodium cooled fast reactor calculations. It is based on nuclear data contained in the Nuclear Data Library and is a 33-group set extending down to 0.4 eV with resonance self-shielding factors automatically calculated in the diffusion theory code CRAM.⁽³⁾

3. Critical Enrichment

The best estimate of the critical enrichment for the first charge of PFR fuel elements is obtained by extrapolation from the assembly ZEBRA 6⁽¹⁾ which is the largest plutonium fuelled assembly so far constructed. Appropriate adjustments may be obtained by correcting for the difference between measured and calculated critical enrichments for ZEBRA 6 and by allowing for differences in the core compositions of PFR and ZEBRA 6. Reactivity worth measurements at the centre of ZEBRA 5J are also relevant because

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the central region of this assembly had a composition similar to that of the PFR core. Detailed consideration of critical mass calculations show that when uncertainties in temperature coefficients are included the fissile material loading corresponding to any state of operation of the PFR can be predicted with an accuracy of about $\pm 4\%$. By the time the first fuel charge for PFR has to be specified it is probable that this uncertainty will be halved.

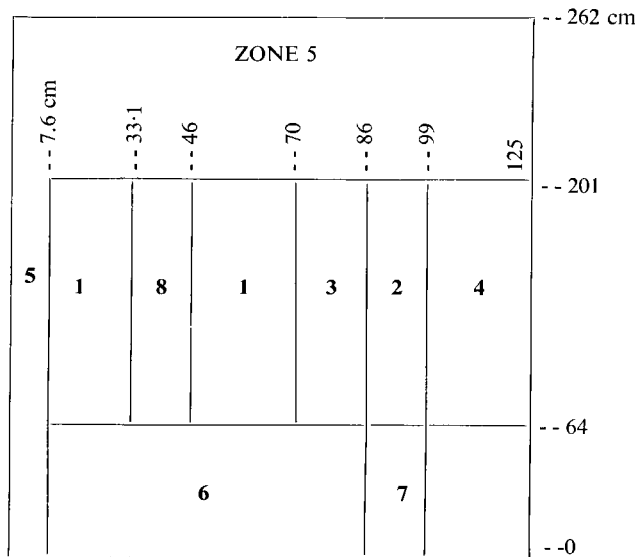


FIG. 1. Layout of PFR calculational model.

4. Control Rod Worths

Only a limited number of materials possess a sufficiently large capture cross-section at high neutron energies to be useful as control material in fast reactors, and this number is further reduced when considerations of cost and availability are introduced. The two materials on which attention has been concentrated for PFR are boron, enriched in the B^{10} isotope, and tantalum.

In general, agreement between calculation and experiment is poor for the reactivity worth of small samples of B^{10} and Ta in fast assemblies. In particular for ZEBRA Assembly 5J, where the spectrum was similar to the PFR, the B^{10} perturbation cross-section was underestimated by 20% and the Ta cross-section was underestimated by 30% when compared with calculations based on the FD2 data set. These errors are discussed further in ref. (1).

To assist the prediction of control rod worths for PFR a series of experiments was undertaken in the second core of ZEBRA utilizing not only small samples, but also substantial rods of absorbing materials. Materials considered were B^{10} (about 90% enriched), natural boron, and tantalum. The

rods extended over the core length and measurements were made at different lattice positions in the core. A summary of the rod data together with experimental and calculated rod worths is given in ref. (1), where it is shown that calculation (five groups, diffusion theory using the Yiftah, Okrent, and Moldauer data) considerably underestimated the reactivity controlled by the rods, the discrepancy varying from about 10% to 30%, and being least for the rods with the most self-shielding. The last observation led Ingram⁽⁴⁾ to suggest that a semi-empirical estimate of a control rod worth could be obtained by measuring the worth of a full sized representation of the rod at the centre of a low power critical assembly, and, using calculated axial and radial scaling factors appropriate to small samples, adjusting the result for off central rod positions.

The basis of this method has been examined by Gubbins⁽⁵⁾ who concluded that errors in the empirical method were partially self-compensating but not in a way which could be justified theoretically. The method offers a useful guide to control rod worth, but a more rigorous treatment is desirable and improved methods (such as a Monte Carlo perturbation technique) are being investigated. Experiments on full size and quarter size control rods are included in the ZEBRA programme for the PFR simulation. These experiments will include interaction measurements and reaction rate scans near and between absorbing rods and the results will be compared with calculations using existing and newly developed techniques.

5. Radial Blanket Design

The radial blanket of PFR consists of one row of subassemblies containing a mixture of graphite and steel, one row of subassemblies containing UO_2 , and two rows of subassemblies containing graphite. This arrangement was selected, as giving a good core radial power form factor, on the basis of earlier studies relating to a 500 MW(e) system. In this work⁽⁶⁾ the properties of various arrangements of blankets were calculated using one-dimensional diffusion theory and the Hansen and Roach 16-group cross-section set⁽⁷⁾ which, at that time, gave the best representation of the low energy and of the spectrum and included resonance self-shielding. The results indicated that the introduction of a layer of moderating material adjacent to the core led to a reduction in critical mass and to an improvement in the core radial power form factor. Breeding, however, was adversely affected whether considered as chemists breeding ratio (i.e. U^{238} capture/Pu fission) or in terms of the breeding gain parameter⁽⁸⁾ in which the reactivity worths of the higher Pu isotopes are given appropriate weighting.

More recently, the efficiency of the radial breeder arrangement for PFR has been re-examined⁽⁹⁾ using diffusion theory and the FD2 cross-section reference set. Detailed examinations of these and other calculations have

shown that the results are uncertain because with moderating material present a significant number of capture reactions in the radial blanket occur at very low energies. It is expected that in the near future improved calculation methods will be available for the optimization of the radial blanket. With these reservations in mind the results from a recent study are presented in Table 1.

TABLE 1. ENRICHMENT, BREEDING, AND POWER FORM FACTOR FOR VARIOUS RADIAL BREEDER ARRANGEMENTS

Case no.	Critical ratio Pu/Pu + U		Internal breeding gain			External radial B.G.	Total B.G. less axial breeder	Radial power form factor
	Inner core	Outer core	Inner core	Outer core	Total			
1	0.170	0.227	-0.124	-0.381	-0.505	0.253	-0.252	1.18
2	0.176	0.235	-0.150	-0.332	-0.482	0.463	-0.021	1.29
3	0.176	0.235	-0.149	-0.337	-0.486	0.331	-0.155	1.28
4	0.172	0.229	-0.135	-0.361	-0.496	0.300	-0.196	1.21

Note. Case 1. 5 in. graphite/steel zone + 5 in. UO_2 breeder.

Case 2. 10 in. UO_2 breeder.

Case 3. 5 in. UO_2 breeder.

Case 4. $2\frac{1}{2}$ in. graphite/steel zone + 5 in. breeder.

The results show that the best breeding is obtained without the moderating zone (case 2). Comparing this case with the reference case (case 1) and one with a half thickness reflecting zone (case 4) it is clear that the latter is much closer to case 1 than case 2. This implies that even a small thickness of moderator is sufficient to reduce significantly the breeding and form factor, the latter being the ratio of peak to average power. Although the present PFR radial blanket is not that which maximizes breeding gain the design of the reactor is such that the graphite elements may be replaced by fertile subassemblies if desired.

In an economic assessment of a power reactor, power form factor, breeding gain, and blanket dimensions are parameters to be optimized and by studying their interaction guidance may be forthcoming as to the theoretical and experimental developments that are necessary. Such aspects were considered in ref. (6) and in ref. (9). The treatment has been further extended⁽¹⁰⁾ using present worth techniques to reduce to a comparable point in time the financial transactions involving the sale of plutonium, the sale of energy produced by the blanket, the cost of fuel fabrication and reprocessing which occur at different points in time.

Direct experimental evidence on the effects of blanket design variations such as have been discussed above is not readily available. However, the

opportunity was taken in ZEBRA Assembly 4A⁽¹¹⁾ to mockup in a 120° sector of the radial reflector a system containing respectively moderator, breeder, and further moderator. The improvement in radial form factor and reduction in critical mass were confirmed. Further experiments are planned in the ZEBRA Assembly which is designed to simulate PFR. The results from these experiments will be used to test the improved theoretical methods and data referred to above.

6. Spatial Distribution of Reaction Rates

Comparison between calculation and experiment on ZEBRA and ZPR assemblies gives confidence that the data and methods in current use are adequate for predicting the general nuclear performance of PFR, with sufficient detail and accuracy for present design purposes. Nevertheless, after commissioning, operational problems will arise which will demand more detailed knowledge of specific reactor conditions, for example the local distortion in reaction rates caused by individual control rod movements and the effect of asymmetries in loading caused by the use of experimental fuel elements. These effects, although secondary, are difficult to calculate. Ideally three-dimensional codes are required, able to handle detailed geometric description and many neutron energy groups. In practice this cannot be achieved with present computer storage. With the increasing emphasis on the fine structural variation in reaction rate distribution, the local discontinuities in reactor composition (and the soft neutron energy spectrum of PFR) it may be necessary to use more rigorous but more complex transport theory codes. The problem will be a formidable one. Comparisons of calculation and experimental results on DFR have shown that to obtain the general distribution of reaction rates the simple two-dimensional diffusion theory programme requires the reactor to be specified in at least fifteen zones with 400 mesh points. The results of this basic calculation have to be modified to accommodate the effects referred to above. By way of illustration, the effect of a single control rod movement (a perturbation upsetting radial flux symmetry) and the effect of the axial movement of a curtain of six control rods both showed local changes in reaction rates of 10–30%. These calculations were made on simplified models; too few energy groups or mesh points were used and one dimension was missing. To remove these limitations would lead to much longer computing times yet the results of these calculations show that significant effects exist so that improved methods will ultimately be required for PFR and indeed are desirable now for DFR. The development of three-dimensional techniques is currently being studied using three approaches, namely Monte Carlo, three-dimensional finite difference approach, and two-dimensional finite difference techniques in conjunction with flux synthesis in the remaining dimension.

Sufficient experimental data to check adequately the calculations involving complex geometry is sparse. Some data are becoming available following post-irradiation examination of DFR elements but at present the experimental accuracy is not high. The much greater accessibility and flexibility of low power critical assemblies enable detailed measurements to be made in complex geometry and it is proposed to investigate reaction rate distributions in detail in the simulated PFR loading in ZEBRA 7. Some asymmetry will be included in the experimental arrangement and this will provide an opportunity for assessing the degree of permissible simplification in the calculation model in order to achieve a specified precision.

7. Doppler Coefficient

The Doppler effect is important to reactor design in two ways. Provided the magnitude of the Doppler effect is large enough to exceed a threshold value the energy release following a nuclear excursion is very substantially

TABLE 2. DOPPLER CONSTANTS ($\partial\partial k/\partial\theta \times 10^{-2}$) FOR 600 MW(th) REACTOR WITH AND WITHOUT 20 VOL. % BeO IN THE INNER ZONE

Case		U ²³⁸	Pu ²³⁹	Total
A (no Be)	With coolant	-1.22	0.27	-0.95
	Without coolant	-0.67	0.15	-0.53
B (Be substituted for coolant)	With coolant	-2.11	0.48	-1.64
	Without coolant	-1.63	0.35	-1.29
C (Be substituted for U ²³⁸)	With coolant	-1.94	0.53	-1.42
	Without coolant	-1.39	0.35	-1.03

reduced.⁽¹²⁾ Secondly, the Doppler effect constitutes the largest component of the power coefficient in large ceramic fuelled fast reactors. The current assessment of the Doppler effect provides confidence that the threshold needed for relief in a nuclear incident is exceeded and once this has been established, its precise magnitude is unimportant. A knowledge of the power coefficients is required in order to specify the amount of reactivity control to be provided. Again high precision is not required. For PFR a generous margin has been allowed by providing sufficient reactivity control to meet an upper estimate of the power coefficient.

Although for PFR it was not considered necessary to take special steps to augment the Doppler effect some calculations have been made with BeO

added to the reactor fuel.⁽¹³⁾ This followed the description of a reactor⁽¹⁴⁾ in which moderator was included to soften the spectrum and so increase the Doppler effect. Addition of moderator material does not have a very large effect on the fissile material density required for criticality so that to maintain the same reactor power density it is necessary to displace U^{238} to accommodate the moderator. Both the addition of BeO and the removal of the necessary amount of U^{238} affect the power coefficients. The penalty incurred by the use of moderating material is that the breeding is reduced. For this reason the use of BeO was confined to the inner (low) enrichment zone (where the sodium coefficient is most positive). The effect of the BeO is summarized in Table 2. The substitution of BeO for coolant (case B—case A) leads to a substantial increase in the Doppler constant. The Doppler temperature coefficient varies inversely with the absolute temperature so that the product $(\theta \cdot \partial k / \partial \theta)$ is a constant—the Doppler constant. Case C represents the more realistic example where the BeO is substituted for U^{238} . In this case the spectrum is softened still further by the removal of U^{238} but this is more than offset by the reduction in U^{238} density. The net effect is that case C provides a smaller increase in the Doppler constant.

More recent calculations, supported by experimental results, indicate that a more probable value for the Pu contribution is negligible so that the U^{238} component will also be the total.

8. Sodium Coefficient

Using diffusion theory, calculations have been made for the model of the PFR using the composition given in Table 3. The sodium reactivity effect is important in two ways. The variation of reactivity with sodium density (and therefore coolant temperature) provides a component of the power coefficient which may be positive. Secondly the total expulsion of some of the sodium from the core could lead to a substantial reactivity addition. Studies have shown that neither of these effects is a significant embarrassment to the reactor designer. The sodium temperature coefficient, even if positive, is heavily outweighed by the Doppler coefficient and no plausible mechanism has been identified which could expel the coolant from a substantial volume of the core except on a time scale which is ample for remedial measures to be applied. Nevertheless, as with the Doppler effect, calculations have been carried out to see whether the sodium coefficient is susceptible to modification by changes in reactor design or fuel composition.⁽¹³⁾

The addition of BeO to the fuel produces a softer neutron energy spectrum and renders the spectrum less dependent on sodium density, reducing the moderating and sodium capture components of the coolant void effect. Independently the removal of U^{238} to make room for the BeO results in fewer fissions above 1 MeV and fewer non-fissile captures at low energies.

Each of these changes will contribute a negative component to the sodium density coefficient (and thereby the temperature coefficient). The effects are summarized in Table 4.

Central perturbation cross-sections provide a useful comparison because

TABLE 3. ZONE DIMENSIONS AND COMPOSITIONS FOR A MODEL OF A PFR. ATOMS/CM³ ($\times 10^{-24}$)

Zone	No.	Pu + U	O	Na	SS	Ni	C	Ta
Core and axial breeder	1	0.00687	0.01374	0.00946	0.0191			
Radial breeder	2	0.0130	0.0260	0.00462	0.0189			
Core radial reflector	3			0.00462	0.0175	0.00822	0.0402	
Radial breeder reflector	4			0.00132	0.0070		0.0721	
Central and top region	5			0.0154	0.0262			
Bottom region (core)	6			0.011	0.0262			
Bottom region (radial breeder)	7			0.00462	0.0192			
Control (out)	8			0.0185	0.0105			
Subass. (in)				0.0118				0.0191

TABLE 4. SODIUM WORTHS FOR DIFFERING VOLUME FRACTIONS OF COOLANT, BeO AND U²³⁸, IN THE INNER ZONE OF A 600 MW(th) FAST REACTOR

Case	Volume fractions				Reactivity change	Central pertn. cross-section
	Coolant	Pu	U ²³⁸	Be	Total loss of sodium	
A	0.420	0.053	0.314	—	$+0.11 \times 10^{-2}$	5.2×10^{-30}
B	0.311	0.055	0.326	0.095	-0.33×10^{-2}	4.1×10^{-30}
C	0.406	0.052	0.234	0.095	-0.63×10^{-2}	3.2×10^{-30}
D	0.406	0.052	0.234	—	-0.50×10^{-2}	3.9×10^{-30}

Note. 10% BeO in 38% fuel represents 3 atoms of Be to 2 atoms of U²³⁸.

these are mainly determined by moderation and resonance capture effects in the central zone. Assuming that the sodium effect is smaller than the effect from uranium and beryllium, the compositional changes case C—case A and case C—case D are approximately equivalent to the addition of about 10% beryllium. The numerical differences between them are also similar (1.1 and 0.7×10^{-30}). Similarly case A—case D (1.3) and case B—case C (0.9) are primarily changes in U²³⁸ density, but with a secondary effect from a

change in coolant density in the latter case. The results are consistent with the supposition that the addition of beryllium has numerically the same effect as the loss of the U^{238} that it displaces. The displacement of sodium has a smaller effect. A similar analysis is possible on the total loss effect of Table 4. It is more complex as the compositional changes were confined to the central zone, but examination of the detailed components has led to a similar conclusion to that given for the central perturbation.

9. Effect of Burnup on Reactivity Coefficients

During the operational life of a fast reactor changes will occur in the fuel composition in both core and blankets due to burnup, fuel recycling and possibly changed sources of make-up plutonium. So far as core fuel is concerned, if plutonium is recycled, then the equilibrium isotopic composition eventually reached in the fuel depends only on the quality of the make-up plutonium supply. On the assumption that the initial charge for PFR is made from magnox reactor 3000 MWd/t plutonium, then if fuel is recycled and make-up plutonium is from fast reactor blanket material, the long-term Pu^{240} content of the plutonium would rise to about 30%. If, however, magnox material was used as make-up, the Pu^{240} content would rise to about 40%.

Such long-term changes in plutonium composition will affect a number of important characteristics of the system, such as the fuel enrichment required, the rate of reactivity loss during irradiation, the breeding gain, and the temperature and power coefficients of the reactor, notably those associated with sodium removal and the Doppler effect.

The variation in feed enrichment resulting from changes in plutonium isotopic composition is not, in itself, cause for great concern, although it may lead to complications in manufacturing and accounting processes. Such changes are gradual and some freedom exists in core loading arrangements because of the two enrichment zone designs and the constant pitch and dimensions of all subassemblies. Nor is the effect on reactivity required for burnup control likely to be greatly affected. Of more importance are the effects on the amount of reactivity required to take the system to power and the degree of self-protection which is available from the reactivity coefficients under accident conditions.

Fuel composition changes have a significant effect on the Doppler coefficients and the reactivity change caused by loss of coolant. Macdonald⁽¹⁵⁾ has considered the reactivity effects of sodium voiding in detail for two models, the first, containing fuel with 17% Pu^{240} and no fission products, and the second, a geometrically similar system with heavy fuel atoms reduced by 10%, the appropriate number of fission products added and with the Pu^{240} isotope increased to 30%. The various contributions to the total reactivity

change on voiding all core subassemblies are given in Table 5. The large positive contribution arising from the moderation term in the burnup case is a result of the capture in fission products and the high Pu^{240} content.

TABLE 5. REACTIVITY CHANGE ON VOIDING ALL CORE SUBASSEMBLIES (FD2 CROSS-SECTIONS)
 $\delta k/k(\%)$

		Core total	Upper axial breeder	Lower axial breeder	Flow mixing region	Gas expn. space	Totals
Standard core	Capture	0.223	0.007	0.008	0.001	0.001	0.240
	Moderation	1.655	0.138	0.130	0.031	0.132	1.985
	Leakage	-1.947	-0.307	-0.336	-0.075	-0.104	-2.769
	Resonance	0.197	0.012	0.014			0.223
	Total	0.128	-0.150	-0.184	-0.043	-0.071	-0.320
Burnt-up core	Capture	0.202	0.009	0.011	0.002	0.004	0.228
	Moderation	3.049	0.179	0.154	0.050	0.036	3.468
	Leakage	-1.922	-0.350	-0.361	-0.142	-0.171	-2.946
	Resonance	0.122	0.014	0.019			0.155
	Total	1.451	-0.148	-0.177	-0.090	-0.131	+0.905

TABLE 6. VARIATION OF ISOTHERMAL DOPPLER AND SODIUM REACTIVITY COEFFICIENTS WITH BURNUP AND CONTROL ROD POSITION ($\times 10^{-6}/^{\circ}\text{C}$)

Enrichment %		Doppler coefficient total	Core and axial blanket sodium coefficients			
Inner core	Outer core		Absorption	Spectrum	Leakage	Total
1. 17.1	22.8	-8.9	0.81	7.60	-8.05	0.4
2. 18.7	24.9	-6.6	0.73	9.14	-8.19	1.7
3. 18.5	24.6	-6.9	0.68	8.62	-7.61	1.7
4. 20.0	26.7	-5.3	0.63	9.85	-7.62	2.9
5. 21.3	28.5	-6.1	0.72	9.80	-8.13	2.4
6. 24.4	32.5	-5.3	0.67	11.47	-8.07	4.1

1. Full fuel density, no fission products, no control absorber.
2. Fuel density reduced by 5%, and a number density of fission products equal to 10% of the original U and Pu density, no control absorber.
3. Case 1 with control rods inserted half-way.
4. Case 2 with control rods inserted half-way.
5. Case 2 with Pu composition 239:240:241:242 = 63:30:5:2.
6. Case 2 with 40% Pu^{240} .

The total reactivity change on voiding the core and axial regions is -0.32% $\delta k/k$ for the clean core and $+0.9\%$ $\delta k/k$ for the burntup core. The most positive effects are calculated to be $+0.85\%$ $\delta k/k$ for the clean core and

+1.78% $\delta k/k$ for the burntup core, although it should be pointed out that voids of the shapes required to give these results would not be formed under realistic accident conditions. The most positive reactivity gain from the voiding of one subassembly is calculated to be 4×10^{-4} , and it is clear that before significant reactivity changes could occur, simultaneous voiding of a number of subassemblies would have to take place.

Doppler and sodium temperature coefficients of reactivity have been calculated for a number of assumed core conditions. The results for some representative cases are shown in Table 6 and give some indication of the variation to be expected in these coefficients during reactor operation. The inclusion of fission products, control absorber or high Pu^{240} content, reduce the Doppler coefficient and make the sodium coefficient more positive. None of these changes is sufficient to vary the earlier remarks, relating to reactor safety. The reduced Doppler effect is still adequate to afford protection in a core meltdown and the reactivity gain on losing coolant is only 4×10^{-4} per subassembly even in a reactor with high Pu^{240} content and a fully burntup core.

10. Comparison of PFR and DFR Reactivity Coefficients and Kinetics

Predictions of the kinetic characteristics of a reactor at power cannot be checked in critical assemblies of the ZPR or ZEBRA type. It is therefore instructive to examine the PFR kinetics against the background of the experience gained during the operation of the Dounreay Fast Reactor.

11. Isothermal Temperature Coefficients and Power Coefficients

Isothermal temperature coefficients provide a check on the nuclear data, i.e. material worths and Doppler effects, used in calculations of kinetic responses, as opposed to the power coefficients which are, in addition, dependent on temperature distribution. In DFR reasonably good agreement was found between the calculated and experimental isothermal coefficients for the first two (simple) loadings.⁽¹⁶⁾ For later cores power flattening increased the reactivity worth of materials in the inner radial blanket, and agreement became relatively poor. The largest uncertainties arose in the calculation of this blanket contribution which, unlike the core contribution, was found to be very sensitive to the approximations made if few-group calculations were employed or if the geometry was simplified. Sixteen-group Yiftah, Okrent, and Moldauer cross-sections⁽¹⁷⁾ were used in the calculations summarized in Table 7. No satisfactory experimental method was found of separating the components of the blanket contributions, because of uncertainties in the blanket flow distributions and a lack of instrumentation.

It has been confirmed experimentally that the magnitude of the PFR

Doppler temperature coefficient is reasonably well calculated using the FD2 data set and it is substantially negative. When compared to the DFR contributions it is seen that the Doppler effect is relatively more important, while the other effects are much less important. The change to a softer neutron spectrum is responsible for the increased Doppler effect, and also for increases in the capture and spectral components of the coolant effect (which tend to cancel the leakage effect and so give a small total coolant

TABLE 7. ISOTHERMAL TEMPERATURE COEFFICIENTS ($\times 10^{-6}/^{\circ}\text{C}$)

	Calculated contribution					Total	
	Core fuel	Core coolant	Core structure	Total Doppler	Axial and radial blanket	Calculated	Measured
DFR* PER	-5.8 -6	-9.0 +0.4	-3.9 -1.9	+0.4 -8.0	-11.7 -2.0	-30 -17.5	-47 $\pm$ 1 —

* DFR in this section refers to charge B4 which was a two zone core loading and a natural uranium blanket and was the first charge run at full power.

TABLE 8. REACTIVITY COEFFICIENTS ESTIMATED FOR PFR AND DFR CORE AT FULL POWER and Flow ($\times 10^{-6}/\text{MW}$)

	Contributor					
	Fuel	Structure	Coolant	Doppler	Bowing	Total core
PFR	-1.5	0	-0.3	-5.0	-0.2	-7.0
DFR (small disturbances)	0	-6	-15	+2	+9	-10
DFR standard set	-22	-6	-15	+2	+9	-32

effect). The decrease in the height/diameter is responsible for the decrease in the relative importance of the structure and blanket contributions. The isothermal temperature coefficient will be measured during the commissioning of PFR and the decrease in the relative importance of the blanket is expected to lead to better agreement between calculation and experiment.

The power coefficient is derived from the temperature coefficient and the average temperature rise per megawatt. Although the temperature coefficient may be well predicted the temperature distribution within the fuel pin remains as an uncertainty in determining the power coefficients. Metallurgical changes during fuel life will also have an effect on the mean fuel temperature so that the Doppler component of the power coefficient may vary with the mean

irradiation life of the core charge. It is usual to express burnup behaviour in terms of $\int k d\theta$. Although this is sufficient for metallurgical purposes it does not provide information on the temperature distribution within the fuel. A number of measurements of the thermal conductivity of unirradiated fuel have been made but few data are available from irradiated fuel.

Table 8 gives the estimated components of the power coefficients for the PFR and the DFR. For the DFR two sets are quoted. The first set, for small disturbances, is believed to be applicable when power changes of the order of 1 MW or less occur. The second set gives the coefficients estimated for power variations of more than a few megawatts. The difference between the sets lies in the magnitude of the axial fuel expansion which is small for small excursions in power. The boundary between the two sets is not well defined, but oscillator measurements⁽¹⁷⁾ which typically produce a high frequency amplitude of 0.5 MW, at 60 MW, tended to confirm the total of the first set, while negative ramp changes⁽¹⁸⁾ which cause changes of 5–10 MW, confirmed, approximately, the total of the standard set. An explanation for the non-linear nature of the axial fuel expansion has been based on the elastic properties of a column of pellets which touch at highly stressed points of contact. Such a column exhibits mechanical hysteresis when it is cyclically stressed, the initial strain being taken on a limited contact area.⁽¹⁹⁾

Similarly, non-linear effects may be expected with the expansion of the oxide fuel of PFR. However, the fuel expansion effect is only 20% of the total core power coefficient in PFR, whereas in DFR (standard set) it is 70%. This is one consequence of the importance of the Doppler contribution to the PFR coefficient. The Doppler contribution in PFR will vary with burnup, mainly because of spectrum changes caused by build-up of fission products and higher plutonium isotopes, but these changes are more predictable than those associated with the axial expansion of fuel. Although the power coefficient in PFR will be non-linear because of the approximately T^{-1} dependence of the Doppler effect, sharp non-linearities such as that observed between the two sets of power coefficients of DFR are unlikely. Thus oscillator measurements on PFR using small amplitude reactivity changes are likely to be of more use in predicting the responses for large changes than they were in DFR.

12. Slow Response Feedback and Steady State Measurements

Another feature of the DFR is the comparatively large amplitude of feedback whose time response is slower than the main core contributors. A total of about -4 milliniles/MW out of -7 milliniles/MW was found, covering a wide spectrum of response times from about 5 to 150 sec, associated with expansions in the blanket zones. This is not including the feedback via heat exchangers and the inlet temperature. Some separation of these slow effects

was achieved both with the oscillator and by means of the ramp reactivity measurements, but the potentially very useful steady state measurements of reactivity at differing coolant flow rates and power did not achieve a useful separation because it was not possible to vary core and blanket flow completely independently. In PFR, although the core and blanket coolant flows are not separable, the magnitude of the blanket contribution is comparatively small (approximately 10% of the total feedback). Thus steady state measurements in PFR will provide useful information on the components of the core coefficient.

13. Problems of Measurement of Kinetic Behaviour

The kinetic behaviour of DFR has been measured using the reactivity oscillator, reactivity ramps, and coolant temperature noise. Difficulties were encountered in all measurements and although some of them were specific to the particular design of DFR it is probable that PFR will pose similar problems.

A reactivity oscillator must have a constant reactivity amplitude over a wide range of rotational speeds, freedom from backlash and freedom from movement along a core radius. This exacting specification has to be met over the full power range in spite of the high temperature sodium environment. In addition, in DFR, twenty-four coolant circuits return coolant at widely differing temperatures (10°C) to a common inlet header tank where mixing is incomplete so that the inlet temperature has a large ($\frac{1}{2}$ °C) noise amplitude. Low frequency disturbances of larger amplitude are also present.

Measurements of DFR reactor noise included auto- and cross-correlation of thermocouple and flux signals. The results were difficult to interpret because the thermocouple response times were comparatively long and considerable noise amplitude existed in the coolant inlet temperature. Nevertheless, this technique did provide an estimate for the lumped time response of the core (0.4 ± 0.2 sec).

Reactivity ramp measurements, in conjunction with a small analogue computer, have successfully demonstrated the presence of stabilizing feedback. Although giving the magnitude it is possible to give only a limit to the time constant of the feedback. For DFR it has been shown that the fast feedback acts in less than about 1 sec (calculated time constant 0.4 sec) and has a magnitude which agrees with the calculated value within the experimental error of 20%. The advantage of the reactivity ramp is that an installed control rod may be used, and the reactor time required is very short.⁽¹⁸⁾

It is proposed to build a special control rod in PFR which will be capable of producing sinusoidal, square wave or pseudo random reactivity changes as required. The accuracy of these methods of measuring the reactor response will be limited primarily by the reactor power noise. The major source of

this noise will again probably be associated with the inlet temperature fluctuations caused by the mixing of separate inlet coolant streams although closer control of coolant inlet temperature is specified than was possible in DFR.

For the several reasons previously discussed the interpretation of the kinetic measurements is likely to be simpler in PFR than it is in DFR. The most important reasons can be summarized as follows:

- (i) Measurements with small amplitude reactivity changes (sinusoidal responses, noise measurements, etc.) will provide information applicable to transients of larger amplitude. This is because in PFR the largest component of the power coefficient will be the Doppler effect whereas in DFR the largest component comes from the fuel expansion and this varies in a strongly non-linear manner with the amplitude of the power perturbation.
- (ii) The slowly responding blanket regions will be relatively unimportant in PFR, and steady state measurements of feedback will provide some information on the important fast-acting core components.

The kinetic measurements on PFR should also be simplified by the much closer control on coolant inlet temperature. Improved experimental conditions are also expected from instrument development and the possible inclusion of on-line computation.

14. Fault Studies

The kinetic response of a reactor to transient changes in reactivity and coolant flow rate are generally complex but may be considerably simplified

TABLE 9. CALCULATED SIMPLE MODEL TIME
CONSTANTS AT FULL COOLANT FLOW RATE
(SECONDS)

	Fuel	Coolant	Flow transit time
DFR	0.37	0.37	0.12
PFR	1.2	1.2	0.15

for survey purposes because three widely differing time scales may be identified, associated with reactor core, delayed neutrons, and effective reactor period. If the response time of the core is sufficiently fast the fuel and coolant temperatures may be considered to remain in equilibrium with the instantaneous values of power and coolant flow rate. Under these conditions considerable simplification of the kinetic equations is possible permitting analytic

solutions for the response of fuel, can, and coolant temperatures to ramp and step reactivity additions and coolant flow transients.^(20,21) The reactor time constants for PFR are short (Table 9) and general surveys have been carried out using these methods.

The current emphasis in kinetics analysis is tending towards evaluation of accident conditions⁽²²⁾ which relate to the specific reactor design. Studies have been undertaken to determine the reactor behaviour following seizure of a coolant pump or failure of a duct. Precision is not necessary for these calculations because accident conditions are not clearly defined.

15. General Conclusion

Physics data for the design of the PFR have been provided with an accuracy which is adequate for present design purposes. It is anticipated that higher precision and more detailed information will be required for operational purposes on the PFR and for the design of subsequent fast reactors. Development of methods and data are in progress and theoretical results will be checked against experimental results from low power assemblies and, ultimately, from PFR itself.

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